



CSE387 (UG2023) - Data Mining and Business Intelligence (39678)

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Overview:

Breast cancer diagnosis is extremely time-critical. Using machine learning, we aim to classify tumors as:

Label	Meaning
0	Benign (non-cancerous)
1	Malignant (cancerous)

Dataset: Breast Cancer Wisconsin (Diagnostic)

Records: 569 patients

Features: 30 numerical tumor characteristics

Goal: Build the most accurate machine learning model for diagnosis support.

Preprocessing and Feature Scaling:

Before model training, we ensured the data was ready for machine learning:

- Removed ID column
- Converted diagnosis B $\rightarrow$ 0, M $\rightarrow$ 1
- No missing values or duplicates detected

- We used RobustScaler instead of StandardScaler as medical data contains real outliers due to tumor abnormality

Scaling Method	Effect on Outliers
StandardScaler	Outliers weaken scaling & distort results
RobustScaler	○ Less affected by outliers ○ Reduces influence without removing data

- Better performance especially for distance-based models like KNN.

DBSCAN Attempt - Unsupervised Learning:

We tested DBSCAN to discover hidden structure in tumor data.

It was rejected as:

- Classified many samples as noise
- Could not separate malignant vs benign clearly
- Only 2–5 features showed acceptable clustering
- Dataset requires a supervised approach therefore we proceed to classification.

Feature Selection Strategy – to boost accuracy

We tested three different input feature sets:

Feature Group	Method	Features Used
1) All features	Original dataset	30
2) Top 10 correlated features	Pearson correlation with diagnosis	10
3) Top 10 RF important features	Random Forest feature importance	10

Why reduce features:

- Removes noisy / irrelevant components
- Faster training
- Improved accuracy in most classifiers

Models Trained – Supervised Learning:

We trained 6 classification models using each of the 3 feature sets:

Models

- Support Vector Machine (SVM)
- Random Forest Classifier
- Decision Tree
- Logistic Regression
- Naive Bayes
- K-Nearest Neighbors (KNN)

Total evaluations = 6 models × 3 feature sets = 18 experiments

Model evaluation:

KNN, Decision Tree & Naive Bayes

Three supervised machine learning models were applied to classify tumors as Benign (0) or Malignant (1). Each model handles data differently, so performance varies.

1) K-Nearest Neighbors – KNN

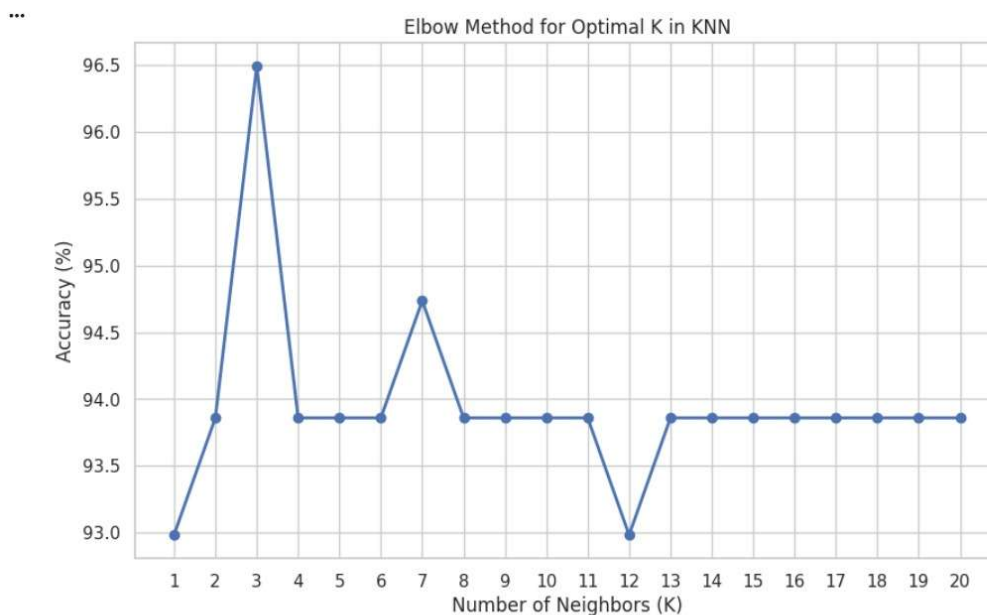
KNN is a classification algorithm that makes predictions based on the closest, most similar cases (neighbors) in the dataset.

This model is well-suited to breast cancer diagnosis because patients with similar tumor features usually belong to the same class.

To determine the best number of neighbors K, we evaluated accuracy across a range of K values (1–20).

The following chart shows how accuracy changes as K increases.

A strong rise in accuracy is observed until K = 3, after which improvements flatten.



Optimal K found using Elbow Method: K = 3

- final chosen value: $K = 3$
- Best performance: 95.61% accuracy
- Very good reliability in diagnosing between benign and malignant tumors

2) Decision Tree Classifier

A Decision Tree splits the data into branches based on feature thresholds, forming if-else rules that mimic medical logic.

- Easily interpretable by doctors
- Shows which tumor characteristics (e.g., area, radius, concavity) contribute to malignancy

Final performance:

Accuracy: 92.98%

Good decision support model, but slightly more prone to overfitting compared to KNN

3) Gaussian Naïve Bayes

Naive Bayes is a probabilistic model that classifies based on likelihood estimates from the features.

- Fast learning
- Low computational cost
- Good baseline model for medical screening

However, breast cancer features are often correlated (e.g., area and radius), which decreases Naive Bayes accuracy on this dataset.

Final performance:

Accuracy: 93.86%

Lower than KNN & Decision Tree, but still useful for quick classification

Experimental Results – Accuracy:

Model	Feature Set	Accuracy	F1-Score (Weighted)	Recall (Weighted)	Support
Logistic Regression	All Features	97.37%	0.97	0.97	114
Random Forest	All Features	97.37%	0.97	0.97	114
Random Forest	Top 10 Corr	96.49%	0.96	0.96	114
KNN	All Features	95.61%	0.96	0.96	114
KNN	Top 10 Corr	95.61%	0.96	0.96	114
SVM	All Features	95.61%	0.96	0.96	114
Logistic Regression	Top 10 Corr	95.61%	0.96	0.96	114
Logistic Regression	Top 10 RF	95.61%	0.96	0.96	114
Random Forest	Top 10 RF	95.61%	0.96	0.96	114
SVM	Top 10 RF	95.61%	0.96	0.96	114
SVM	Top 10 Corr	95.61%	0.96	0.96	114
Naive Bayes	Top 10 RF	93.86%	0.94	0.94	114
Naive Bayes	Top 10 Corr	93.86%	0.94	0.94	114
Decision Tree	All Features	92.98%	0.93	0.93	114
Naive Bayes	All Features	92.11%	0.92	0.92	114
Decision Tree	Top 10 RF	91.23%	0.91	0.91	114
Decision Tree	Top 10 Corr	90.35%	0.9	0.9	114

Final Model Selection:

Logistic Regression using All Features

- Accuracy: 97.37%
- Best performance among all tested configurations
- Strong balance between precision and recall
- Fast and efficient for medical diagnosis

Why Logistic Regression performed best:

- Handles linearly separable medical data very well
- Performs strongly even without feature reduction
- Robust to noise after applying Robust Scaling
- Interpretable for doctors, coefficients explain each feature's importance

Key Findings:

Logistic Regression achieved the highest accuracy overall

- Feature selection improved several models, but not needed for LR
- SVM and Random Forest also showed strong performance
- Naive Bayes was reliable but less accurate
- KNN worked well, but not as accurate as LR in final evaluation

Overall: Logistic Regression is the most trustworthy diagnostic model

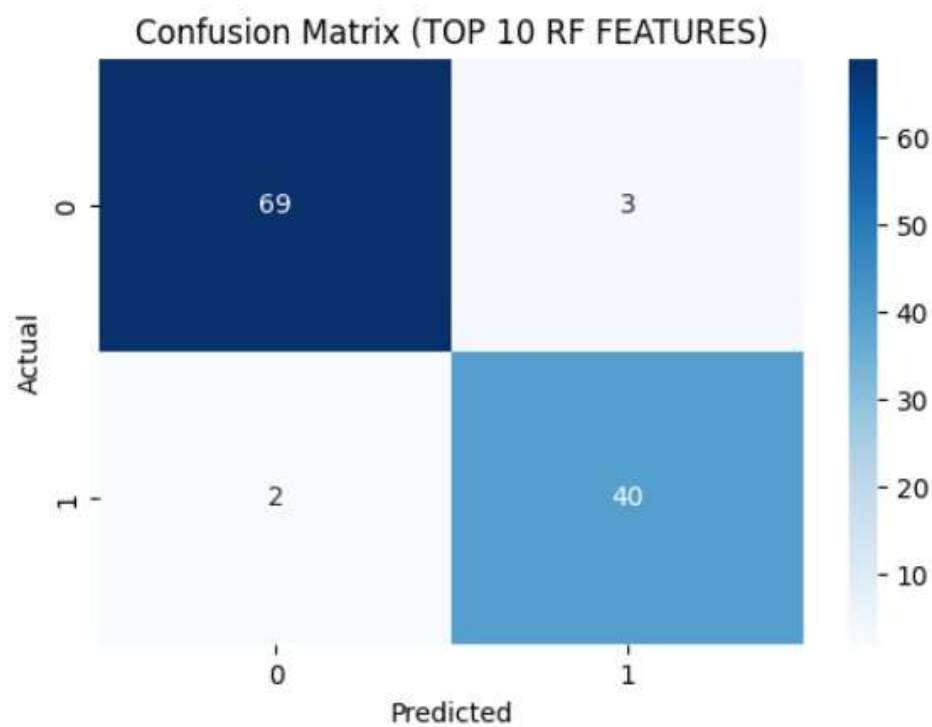
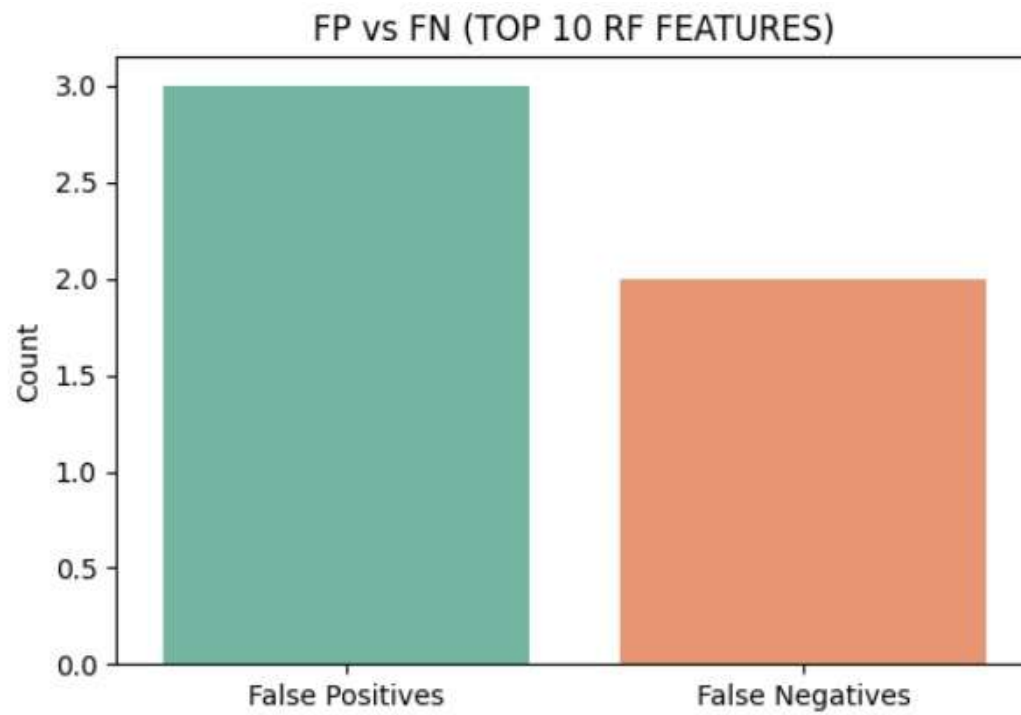
Conclusion:

This study developed multiple machine learning models for automated breast cancer diagnosis.

Through performance comparison, Logistic Regression using all features demonstrated the best accuracy and reliability.

Final Outcome Summary:

Aspect	Outcome
Preprocessing	Improved data quality and fairness across features
Feature Selection	Helped many models, but LR best with full features
Best Algorithm	Logistic Regression
Best Accuracy	97.37%



Notebook links:

- Naïve Bayes :
https://colab.research.google.com/drive/1kugujVGxTyQt55A3AXc5xX3JTatz3_9n?usp=sharing
- KNN:
https://colab.research.google.com/drive/19rxRz3befr5htHLBuzd9AkWnh_ptvSNu?usp=sharing
- Logistic Regression:
<https://colab.research.google.com/drive/188r9H6IqjnHPRoJbG3Tqhi33irdsPFPV?usp=sharing>
- Random Forest: <https://colab.research.google.com/drive/1LoPIVgH-Bo7urT-Vn-DLmbBVaDVtNMYn?usp=sharing>
- SVM:
<https://colab.research.google.com/drive/19kK3vOc9ekqUgDwxNbOo4irtXFzPve95?usp=sharing>
- Decision Tree:
<https://colab.research.google.com/drive/19kK3vOc9ekqUgDwxNbOo4irtXFzPve95?usp=sharing>
- Preprocessing:
<https://colab.research.google.com/drive/11FclWC6haEbieTK8DJ2LrCoixa1qtSF?usp=sharing>